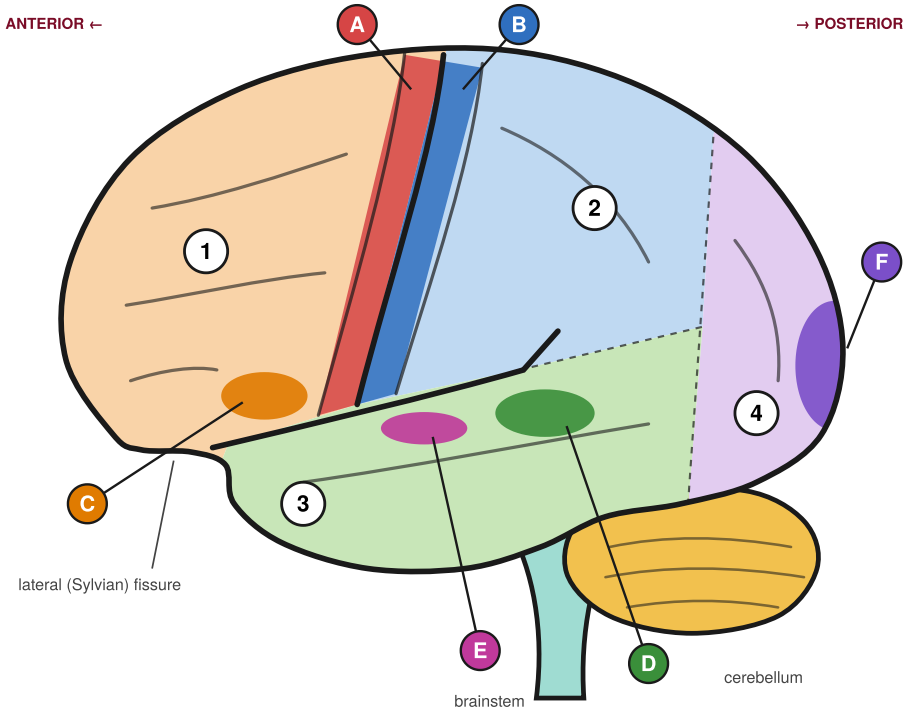
Harvard

Fundamentals of Neuroscience · MicroBachelors® Program

Cerebral Cortex — Lobes & Functional Areas

MCB80x · lateral view · 4 lobes and 6 key cortical areas

DIAGRAM — LEFT HEMISPHERE, LATERAL VIEW



Schematic. Dashed lines are imaginary lobe borders; area sizes and positions are approximate.

Landmarks to know

- **Central sulcus** splits frontal from parietal — motor (A) in front, sensory (B) behind.
- **Lateral fissure** separates the temporal lobe from the frontal and parietal lobes above it.
- Language loop: **Wernicke (D) → arcuate fasciculus → Broca (C)**; usually left hemisphere.

KEY

#	Structure	Function · exam fact
1	**Frontal lobe** Anterior to central sulcus	Planning, decisions, working memory, personality (prefrontal); voluntary movement. **FACT** Phineas Gage: prefrontal damage changed personality and impulse control.
2	**Parietal lobe** Central sulcus → occipital	Touch, body position, spatial attention, combining the senses. **FACT** Right parietal damage → hemispatial neglect of the left side of space.
3	**Temporal lobe** Below lateral fissure	Hearing, language comprehension, object and face recognition; memory (medial). **FACT** Fusiform face area damage → prosopagnosia (face blindness).
4	**Occipital lobe** Posterior pole	Vision: color, motion, form processed in separate areas. **FACT** Each hemisphere sees the opposite visual field.
A	**Primary motor cortex** M1 · precentral gyrus	Sends commands for voluntary movement to the opposite side of the body. **FACT** Motor homunculus: hands and face get oversized territory.
B	**Primary somatosensory** S1 · postcentral gyrus	Touch, pain, temperature, and limb position from the opposite side. **FACT** Sensory homunculus mapped by Penfield's stimulation studies.
C	**Broca's area** Left inferior frontal gyrus	Speech production and grammar. **FACT** Broca's aphasia: slow, effortful speech; comprehension mostly spared.
D	**Wernicke's area** Left posterior sup. temporal	Language comprehension. **FACT** Wernicke's aphasia: fluent but meaningless speech, poor comprehension.
E	**Primary auditory cortex** A1 · Heschl's gyrus	First cortical stop for sound; mostly tucked inside the lateral fissure. **FACT** Tonotopic: neurons arranged by sound frequency.
F	**Primary visual cortex** V1 · calcarine sulcus	First cortical stop for vision; mostly on the medial surface. **FACT** Retinotopic map; input arrives from the thalamus's LGN.

Personal MCB80x study chart — not an official HarvardX/edX publication.